Decision-time information in rolling airline recovery

Schematic AOCC snapshot: similar delay, different operational exposure

Part 1. AOCC decision-time state

Two aircraft-linked chains assessed during turnaround

Predecessor in-block — Successor off-block — Successor take-off

factual — unresolved — unresolved

Delay priority (P^D) similar predicted take-off delay

Chain A TURN

Turnaround opportunity | Aircraft continuity | Downstream exposure

Chain B TURN

Passenger exposure | Connection exposure | Operating-resource burden

Consequence-based priority (P^C)

Flight, Passenger and Resource consequences distinguish these chains

Observed: predecessor status | Infer: successor exposure | Act: shortlist → target

Part 2. Stage I screening

Delay versus consequence-based priority

Which chains enter the limited review shortlist?

Delay priority (P^D) A B

similar delay-based priorities

Consequence (P^C) A B

context may change selection

Same candidate set and capacity; different priority signals

Part 3. Stage II local recovery

Within one shortlisted chain: u=0 or u>0

Compare consequences and effort across feasible targets

Feasible target set — recovery headroom

(u=0): no recovery — (u>0): positive target

Information value depends on the supported decision